2

223611496

8

DNS HTTP

1 DNS

2 HTTP HTTPS

3 HTTP HTTP

1

2 Wireshark DNS DNS

3 Wireshark HTTP/HTTPS HTTP HTTPS TCP/DNS

4 HTTP HTTP

2.3.1 DNS

1 DNS

1.1 “ipconfig /displaydns” 2

```
PS C:\Users\OMEN> ipconfig /displaydns
```

```
Windows  IP

    fe3cr.delivery.mp.microsoft.com
    ----------------------------------------
        . . . . . . . . : fe3cr.delivery.mp.microsoft.com
        . . . . . . . . : 5
        . . . . . . . . : 515576
        . . . . . . . . : 8
      . . . . . . . . . :
    CNAME        . . . . . : fe3.delivery.mp.microsoft.com


        . . . . . . . . : fe3.delivery.mp.microsoft.com
        . . . . . . . . : 5
        . . . . . . . . : 515576
        . . . . . . . . : 8
      . . . . . . . . . :
    CNAME        . . . . . : glb.cws.prod.dcat.dsp.trafficmanager.net


        . . . . . . . . : glb.cws.prod.dcat.dsp.trafficmanager.net
        . . . . . . . . : 1
        . . . . . . . . : 515576
        . . . . . . . . : 4
      . . . . . . . . . :
    A ( )      . . . . : 135.232.92.97


    i1.hdslb.com
    ----------------------------------------
        . . . . . . . . : i1.hdslb.com
        . . . . . . . . : 5
        . . . . . . . . : 528866
        . . . . . . . . : 8
      . . . . . . . . . :
    CNAME        . . . . . : i0.hdslb.com.cdn.dnsv1.com


        . . . . . . . . : i0.hdslb.com.cdn.dnsv1.com
        . . . . . . . . : 5
        . . . . . . . . : 528866
        . . . . . . . . : 8
      . . . . . . . . . :
    CNAME        . . . . . : 427884.sched.sma-dk.tdnsstic1.cn


        . . . . . . . . : 427884.sched.sma-dk.tdnsstic1.cn
        . . . . . . . . : 1
        . . . . . . . . : 528866
        . . . . . . . . : 4
      . . . . . . . . . :
    A ( )      . . . . : 111.51.105.102
```

1.2 “ipconfig /flushdns” “ipconfig /displaydns”

```
PS C:\Users\OMEN> ipconfig /flushdns

Windows IP 配置

已成功刷新 DNS 解析缓存。
PS C:\Users\OMEN> ipconfig /displaydns

Windows IP 配置

    opcenter.example.com
    ----------------------------------------
    没有 AAAA 类型的记录

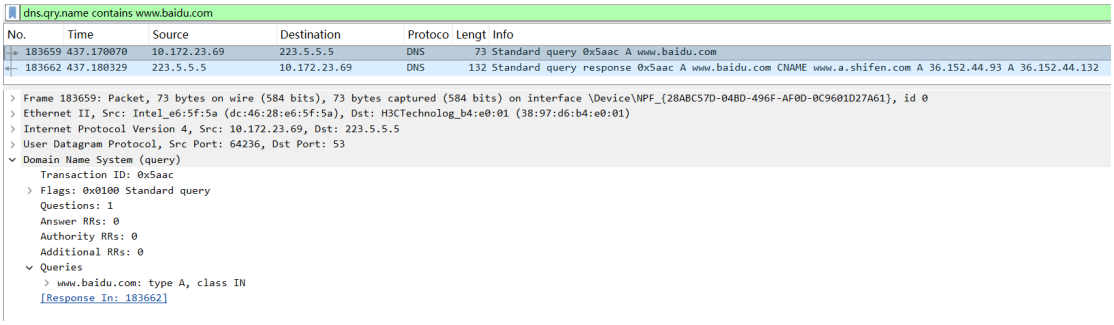

    记录名称. . . . . : opcenter.example.com
    记录类型. . . . . : 1
    生存时间. . . . . : 604800
    数据长度. . . . . : 4
    部分. . . . . : 答案
    A (主机)记录  . . . . : 202.117.21.118


    notebook.example.com
    ----------------------------------------
    没有 AAAA 类型的记录


    记录名称. . . . . : notebook.example.com
    记录类型. . . . . : 1
```

2 DNS Wireshark www.baidu.com DNS

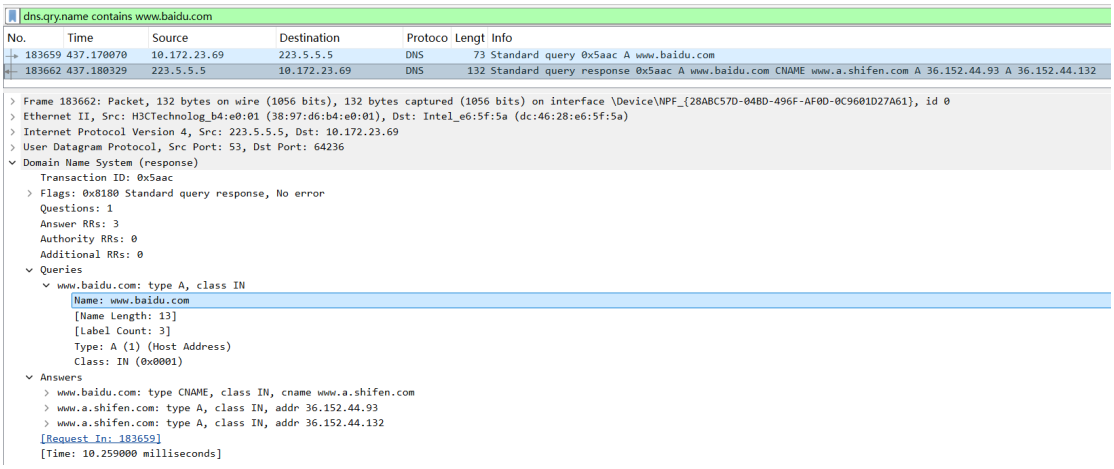
2.1 Wireshark “ ” DNS dns.qry.name contains www.baidu.com



2.2 DNS

UDP TCP	**UDP**
	223.5.5.5
	53
	A
	www.baidu.com

2.3 DNS “ ”



2.4 2.3

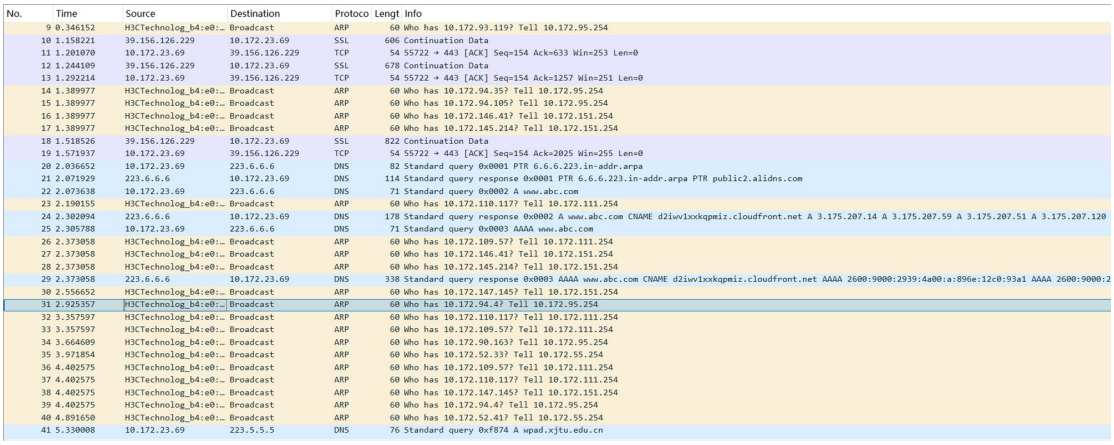
Answers	**3**
Authoritative NS	**0**
Additional	**0**

3 **DNS** “ipconfig /flushdns” Wireshark

DNS **223.6.6.6** X **www.abc.com** nslookup

www.abc.com **223.6.6.6** DNS

3.1 Wireshark “ ”



3.2 DNS

6

3.3 DNS

DNS DNS	
PTR	
www.abc.com	**d2iwv1xxkqpmiz.cloudfront.net**

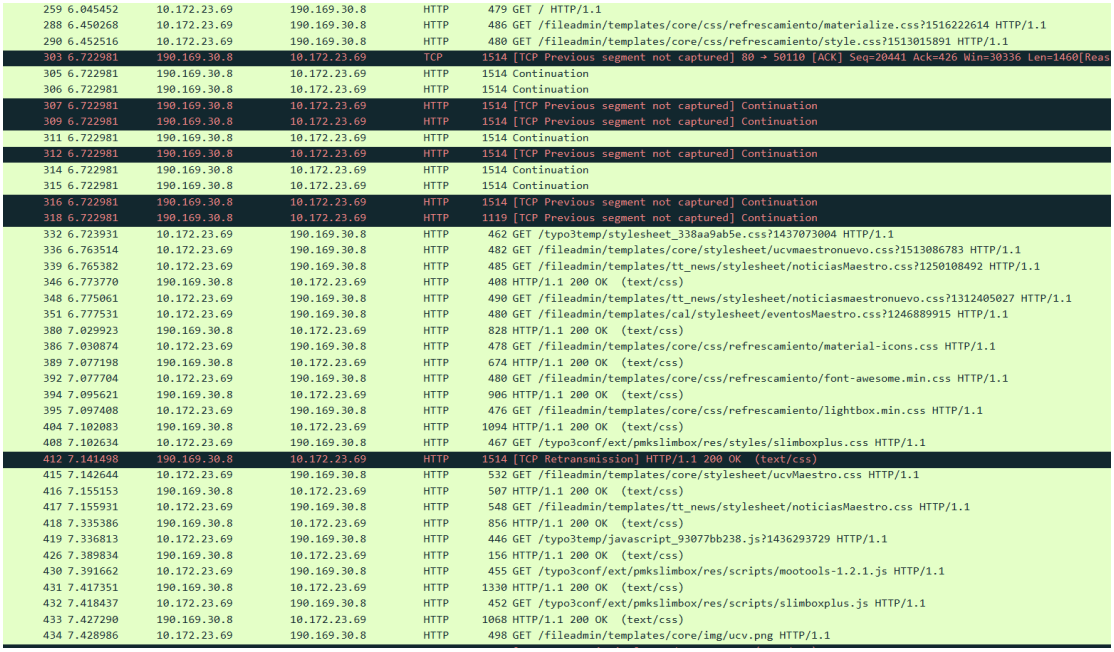
IPv4	**4**

2.3.2 HTTP/HTTPS

4 5 DNS HTTP

4 DNS HTTP

4.1 DNS HTTP HTTP Wireshark HTTP “ ”



4.2

	www.hit.edu.cn	www.customs.gov.mm	www.ucv.ve
IP	202.118.254.135	150.95.82.190	190.169.30.8
RTT	41.263	561.954	303.54 ms
DNS	8	8	4
1 DNS (ms)	8.467	665.077	318.193
(ms)	14322.518	176388.699	19928.113
HTTP	7	16	6
	114	47	124

4.3 HTTP TCP HTTP +“ ”

tcp.stream eq 39

No.	Time	Source	Destination	Protoco	Lengt	Info
252	5.741587	10.172.23.69	190.169.30.8	TCP	66	50110 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
257	6.045127	190.169.30.8	10.172.23.69	TCP	66	80 → 50110 [SYN, ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1460 SACK_PERM WS=128
258	6.045207	10.172.23.69	190.169.30.8	TCP	54	50110 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
259	6.045452	10.172.23.69	190.169.30.8	HTTP	479	GET / HTTP/1.1

4.4 DNS +“ ”+ DNS“ ”

dns and dns.qry.name contains "ucv.ve"

No.	Time	Source	Destination	Protoco	Lengt	Info
237	5.413160	10.172.23.69	223.5.5.5	DNS	70	Standard query 0x6b19 HTTPS www.ucv.ve
238	5.413369	10.172.23.69	223.5.5.5	DNS	70	Standard query 0xff12 A www.ucv.ve
250	5.731353	223.5.5.5	10.172.23.69	DNS	133	Standard query response 0x6b19 HTTPS www.ucv.ve SOA dns1.ucv.ve
251	5.741167	223.5.5.5	10.172.23.69	DNS	86	Standard query response 0xff12 A www.ucv.ve A 190.169.30.8
2250	11.248046	10.172.23.69	223.5.5.5	DNS	78	Standard query 0x7c52 HTTPS ucvnoticias.ucv.ve
2251	11.248241	10.172.23.69	223.5.5.5	DNS	78	Standard query 0x4644 A ucvnoticias.ucv.ve
2252	11.262257	223.5.5.5	10.172.23.69	DNS	94	Standard query response 0x4644 A ucvnoticias.ucv.ve A 190.169.30.46
2807	11.817210	223.5.5.5	10.172.23.69	DNS	129	Standard query response 0x7c52 HTTPS ucvnoticias.ucv.ve SOA dns1.ucv.ve

```
> Frame 251: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF_{28ABC57D-04BD-496F-AF0D-0C9601D27A61}, id 0
> Ethernet II, Src: H3CTechnolog_b4:e0:01 (38:97:d6:b4:e0:01), Dst: Intel_e6:5f:5a (dc:46:28:e6:5f:5a)
> Internet Protocol Version 4, Src: 223.5.5.5, Dst: 10.172.23.69
> User Datagram Protocol, Src Port: 53, Dst Port: 55708
v Domain Name System (response)
    Transaction ID: 0xff12
  > Flags: 0x8180 Standard query response, No error
    Questions: 1
    Answer RRs: 1
    Authority RRs: 0
    Additional RRs: 0
  v Queries
    v www.ucv.ve: type A, class IN
        Name: www.ucv.ve
        [Name Length: 10]
        [Label Count: 3]
        Type: A (1) (Host Address)
        Class: IN (0x0001)
  v Answers
    v www.ucv.ve: type A, class IN, addr 190.169.30.8
        Name: www.ucv.ve
        Type: A (1) (Host Address)
        Class: IN (0x0001)
        Time to live: 3600 (1 hour)
        Data length: 4
        Address: 190.169.30.8
    [Request In: 238]
    [Time: 327.798000 milliseconds]
```

4.5 DNS HTTP 2 +“ ” “ (ms)”

dns and dns.qry.name contains "ucv.ve"

No.	Time	Source	Destination	Protoco	Lengt	Info
237	5.413160	10.172.23.69	223.5.5.5	DNS	70	Standard query 0x6b19 HTTPS www.ucv.ve
238	5.413369	10.172.23.69	223.5.5.5	DNS	70	Standard query 0xff12 A www.ucv.ve
250	5.731353	223.5.5.5	10.172.23.69	DNS	133	Standard query response 0x6b19 HTTPS www.ucv.ve SOA dns1.ucv.ve
251	5.741167	223.5.5.5	10.172.23.69	DNS	86	Standard query response 0xff12 A www.ucv.ve A 190.169.30.8
2250	11.248046	10.172.23.69	223.5.5.5	DNS	78	Standard query 0x7c52 HTTPS ucvnoticias.ucv.ve
2251	11.248241	10.172.23.69	223.5.5.5	DNS	78	Standard query 0x4644 A ucvnoticias.ucv.ve
2252	11.262257	223.5.5.5	10.172.23.69	DNS	94	Standard query response 0x4644 A ucvnoticias.ucv.ve A 190.169.30.46
2807	11.817210	223.5.5.5	10.172.23.69	DNS	129	Standard query response 0x7c52 HTTPS ucvnoticias.ucv.ve SOA dns1.ucv.ve

http and ip.addr == 190.169.30.8

No.	Time	Source	Destination	Protoco	Lengt	Info
259	6.045452	10.172.23.69	190.169.30.8	HTTP	479	GET / HTTP/1.1
288	6.450268	10.172.23.69	190.169.30.8	HTTP	486	GET /fileadmin/templates/core/css/refrescamiento/materialize.css?1516222614 HTTP/1.1
290	6.452516	10.172.23.69	190.169.30.8	HTTP	480	GET /fileadmin/templates/core/css/refrescamiento/style.css?1513015891 HTTP/1.1
305	6.722981	190.169.30.8	10.172.23.69	HTTP	1514	Continuation
306	6.722981	190.169.30.8	10.172.23.69	HTTP	1514	Continuation
307	6.722981	190.169.30.8	10.172.23.69	HTTP	1514	[TCP Previous segment not captured] Continuation
309	6.722981	190.169.30.8	10.172.23.69	HTTP	1514	[TCP Previous segment not captured] Continuation
311	6.722981	190.169.30.8	10.172.23.69	HTTP	1514	Continuation
312	6.722981	190.169.30.8	10.172.23.69	HTTP	1514	[TCP Previous segment not captured] Continuation
314	6.722981	190.169.30.8	10.172.23.69	HTTP	1514	Continuation
315	6.722981	190.169.30.8	10.172.23.69	HTTP	1514	Continuation
316	6.722981	190.169.30.8	10.172.23.69	HTTP	1514	[TCP Previous segment not captured] Continuation
318	6.722981	190.169.30.8	10.172.23.69	HTTP	1119	[TCP Previous segment not captured] Continuation
332	6.723931	10.172.23.69	190.169.30.8	HTTP	462	GET /typo3temp/stylesheet_338aa9ab5e.css?1437073004 HTTP/1.1
336	6.763514	10.172.23.69	190.169.30.8	HTTP	482	GET /fileadmin/templates/core/stylesheet/ucvmaestronuevo.css?1513086783 HTTP/1.1
339	6.765382	10.172.23.69	190.169.30.8	HTTP	485	GET /fileadmin/templates/tt_news/stylesheet/noticiasMaestro.css?1250108492 HTTP/1.1
346	6.773770	190.169.30.8	10.172.23.69	HTTP	408	HTTP/1.1 200 OK (text/css)
348	6.775061	10.172.23.69	190.169.30.8	HTTP	490	GET /fileadmin/templates/tt_news/stylesheet/noticiasmaestronuevo.css?1312405027 HTTP/1.1
351	6.777531	10.172.23.69	190.169.30.8	HTTP	480	GET /fileadmin/templates/cal/stylesheet/eventosMaestro.css?1246889915 HTTP/1.1
380	7.029923	190.169.30.8	10.172.23.69	HTTP	828	HTTP/1.1 200 OK (text/css)
386	7.030874	10.172.23.69	190.169.30.8	HTTP	478	GET /fileadmin/templates/core/css/refrescamiento/material-icons.css HTTP/1.1
389	7.077198	190.169.30.8	10.172.23.69	HTTP	674	HTTP/1.1 200 OK (text/css)
392	7.077704	10.172.23.69	190.169.30.8	HTTP	480	GET /fileadmin/templates/core/css/refrescamiento/font-awesome.min.css HTTP/1.1
394	7.095621	190.169.30.8	10.172.23.69	HTTP	906	HTTP/1.1 200 OK (text/css)
395	7.097408	10.172.23.69	190.169.30.8	HTTP	476	GET /fileadmin/templates/core/css/refrescamiento/lightbox.min.css HTTP/1.1
404	7.102083	190.169.30.8	10.172.23.69	HTTP	1094	HTTP/1.1 200 OK (text/css)
408	7.102634	10.172.23.69	190.169.30.8	HTTP	467	GET /typo3conf/ext/pmkslimbox/res/styles/slimboxplus.css HTTP/1.1
412	7.141498	190.169.30.8	10.172.23.69	HTTP	1514	[TCP Retransmission] HTTP/1.1 200 OK (text/css)
415	7.142644	10.172.23.69	190.169.30.8	HTTP	532	GET /fileadmin/templates/core/stylesheet/ucvMaestro.css HTTP/1.1
416	7.155153	190.169.30.8	10.172.23.69	HTTP	507	HTTP/1.1 200 OK (text/css)
417	7.155931	10.172.23.69	190.169.30.8	HTTP	548	GET /fileadmin/templates/tt_news/stylesheet/noticiasMaestro.css HTTP/1.1
418	7.335386	190.169.30.8	10.172.23.69	HTTP	856	HTTP/1.1 200 OK (text/css)
419	7.336813	10.172.23.69	190.169.30.8	HTTP	446	GET /typo3temp/javascript_93077bb238.js?1436293729 HTTP/1.1
426	7.389834	190.169.30.8	10.172.23.69	HTTP	156	HTTP/1.1 200 OK (text/css)
430	7.391662	10.172.23.69	190.169.30.8	HTTP	455	GET /typo3conf/ext/pmkslimbox/res/scripts/mootools-1.2.1.js HTTP/1.1
431	7.417351	190.169.30.8	10.172.23.69	HTTP	1330	HTTP/1.1 200 OK (text/css)
432	7.418437	10.172.23.69	190.169.30.8	HTTP	452	GET /typo3conf/ext/pmkslimbox/res/scripts/slimboxplus.js HTTP/1.1
433	7.427290	190.169.30.8	10.172.23.69	HTTP	1068	HTTP/1.1 200 OK (text/css)
434	7.428986	10.172.23.69	190.169.30.8	HTTP	498	GET /fileadmin/templates/core/img/ucv.png HTTP/1.1
438	7.457448	190.169.30.8	10.172.23.69	HTTP	1514	[TCP Retransmission] HTTP/1.1 200 OK (text/css)
441	7.460157	10.172.23.69	190.169.30.8	HTTP	504	GET /fileadmin/templates/core/img/historia3.png HTTP/1.1
445	7.479618	190.169.30.8	10.172.23.69	HTTP	1514	[TCP Out-Of-Order] HTTP/1.1 200 OK (text/css)
450	7.646297	190.169.30.8	10.172.23.69	HTTP	621	HTTP/1.1 200 OK (application/javascript)
464	7.739237	190.169.30.8	10.172.23.69	HTTP	1500	HTTP/1.1 200 OK (application/javascript)
511	8.017526	190.169.30.8	10.172.23.69	HTTP	1514	[TCP Out-Of-Order] HTTP/1.1 200 OK (application/javascript)
514	8.026467	10.172.23.69	190.169.30.8	HTTP	507	GET /fileadmin/templates/core/img/autoridades4.png HTTP/1.1
515	8.027460	10.172.23.69	190.169.30.8	HTTP	503	GET /fileadmin/templates/core/img/simbolos.png HTTP/1.1
516	8.028880	10.172.23.69	190.169.30.8	HTTP	506	GET /fileadmin/templates/core/img/patrimonio1.png HTTP/1.1
517	8.029741	10.172.23.69	190.169.30.8	HTTP	463	GET /fileadmin/templates/core/js/refrescamiento/jquery-2.1.4.min.js HTTP/1.1
529	8.084439	190.169.30.8	10.172.23.69	HTTP	692	HTTP/1.1 200 OK (PNG)
537	8.086272	10.172.23.69	190.169.30.8	HTTP	456	GET /fileadmin/templates/core/js/refrescamiento/jquery-ui.js HTTP/1.1

DNS HTTP

5 DNS HTTP

5.1 2-3 Wireshark HTTP

“ ”

No.	Time	Source	Destination	Protoco	Lengt	Info
121	3.238590	10.172.23.69	190.169.30.8	HTTP	534	GET / HTTP/1.1
138	3.593896	190.169.30.8	10.172.23.69	TCP	1514	[TCP Previous segment not captured] 80 → 54713 [ACK] Seq=102
140	3.593896	190.169.30.8	10.172.23.69	HTTP	1514	Continuation
141	3.593896	190.169.30.8	10.172.23.69	HTTP	1514	Continuation
157	3.915674	190.169.30.8	10.172.23.69	HTTP	1514	Continuation
158	3.917967	190.169.30.8	10.172.23.69	HTTP	1514	[TCP Previous segment not captured] Continuation
160	3.917967	190.169.30.8	10.172.23.69	HTTP	1514	Continuation
161	3.917967	190.169.30.8	10.172.23.69	HTTP	1514	Continuation
162	3.917967	190.169.30.8	10.172.23.69	HTTP	1514	Continuation
163	3.917967	190.169.30.8	10.172.23.69	HTTP	1514	Continuation
164	3.917967	190.169.30.8	10.172.23.69	HTTP	1514	[TCP Previous segment not captured] Continuation
166	3.917967	190.169.30.8	10.172.23.69	HTTP	1514	Continuation
167	3.917967	190.169.30.8	10.172.23.69	HTTP	1514	[TCP Previous segment not captured] Continuation
169	3.917967	190.169.30.8	10.172.23.69	HTTP	1514	Continuation
170	3.917967	190.169.30.8	10.172.23.69	HTTP	1514	Continuation
171	3.917967	190.169.30.8	10.172.23.69	HTTP	1514	Continuation
172	3.917967	190.169.30.8	10.172.23.69	HTTP	1514	Continuation
173	3.917967	190.169.30.8	10.172.23.69	HTTP	1514	Continuation
174	3.917967	190.169.30.8	10.172.23.69	HTTP	1514	[TCP Previous segment not captured] Continuation
176	3.917967	190.169.30.8	10.172.23.69	HTTP	1119	Continuation

5.2

	www.hit.edu.cn	www.customs.gov.mm	www.ucv.ve
IP	202.118.254.135	150.95.82.190	190.169.30.8
RTT	41.897	558.214	310.042
DNS	0	0	0
(ms)	773.014	12486.371	679.377
	6	3	1
TCP	1	6	1

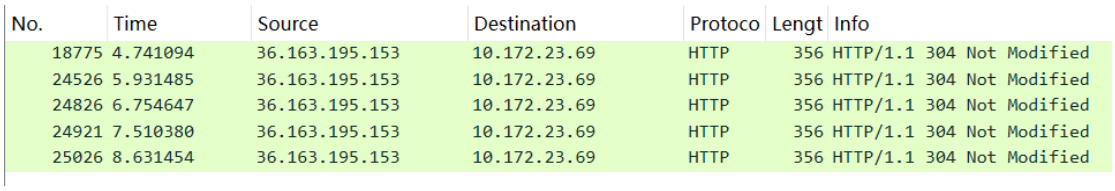
5.3 4 5 4.2 5.2

① DNS DNS
DNS 0 DNS
② HTML CSS JS
TCP
③ DNS

④

6 HTTP

6.1 HTTP **304** "http.response.code == 304" “ ”



No.	Time	Source	Destination	Protoco	Lengt	Info
18775	4.741094	36.163.195.153	10.172.23.69	HTTP	356	HTTP/1.1 304 Not Modified
24526	5.931485	36.163.195.153	10.172.23.69	HTTP	356	HTTP/1.1 304 Not Modified
24826	6.754647	36.163.195.153	10.172.23.69	HTTP	356	HTTP/1.1 304 Not Modified
24921	7.510380	36.163.195.153	10.172.23.69	HTTP	356	HTTP/1.1 304 Not Modified
25026	8.631454	36.163.195.153	10.172.23.69	HTTP	356	HTTP/1.1 304 Not Modified

6.2 304 HTTP

```
> Frame 18680: Packet, 668 bytes on wire (5344 bits), 668 bytes captured (5344 bits) on interface \Device\NPF_{28ABC57D-04BD-496F-AF0D-0C9601D27A61}, id 0
> Ethernet II, Src: Intel_e6:5f:5a (dc:46:28:e6:5f:5a), Dst: H3CTechnolog_b4:e0:01 (38:97:d6:b4:e0:01)
> Internet Protocol Version 4, Src: 10.172.23.69, Dst: 36.163.195.153
> Transmission Control Protocol, Src Port: 49666, Dst Port: 80, Seq: 15604, Ack: 1498263, Len: 614
v Hypertext Transfer Protocol
  > GET / HTTP/1.1\r\n
    Host: www.people.com.cn\r\n
    Connection: keep-alive\r\n
    Cache-Control: max-age=0\r\n
    Upgrade-Insecure-Requests: 1\r\n
    User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0\r\n
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7\r\n
    Accept-Encoding: gzip, deflate\r\n
    Accept-Language: zh-CN,zh;q=0.9,en;q=0.8,en-GB;q=0.7,en-US;q=0.6\r\n
  > Cookie: sso_c=0; sfr=1\r\n
    If-None-Match: W/"69ba9d5f-25df5"\r\n
    If-Modified-Since: Wed, 18 Mar 2026 12:41:03 GMT\r\n
    \r\n
    [Response in frame: 18775]
    [Full request URI: http://www.people.com.cn/]
```

6.3 304 HTTP

```
> Frame 18775: Packet, 356 bytes on wire (2848 bits), 356 bytes captured (2848 bits) on interface \Device\NPF_{28ABC57D-04BD-496F-AF0D-0C9601D27A61}, id 0
> Ethernet II, Src: H3CTechnolog_b4:e0:01 (38:97:d6:b4:e0:01), Dst: Intel_e6:5f:5a (dc:46:28:e6:5f:5a)
> Internet Protocol Version 4, Src: 36.163.195.153, Dst: 10.172.23.69
> Transmission Control Protocol, Src Port: 80, Dst Port: 49666, Seq: 1498263, Ack: 16218, Len: 302
v Hypertext Transfer Protocol
  > HTTP/1.1 304 Not Modified\r\n
    Date: Wed, 18 Mar 2026 13:06:59 GMT\r\n
    Content-Type: text/html\r\n
    Connection: keep-alive\r\n
    Last-Modified: Wed, 18 Mar 2026 12:41:03 GMT\r\n
    ETag: "69ba9d5f-25df5"\r\n
    Age: 29\r\n
    x-ws-request-id: 69baa373_PS-XIY-016qy38_63841-43370\r\n
    x-via: 1.1 PS-XIY-016qy38:33 (Cdn Cache Server V2.0)\r\n
    \r\n
    [Request in frame: 18680]
    [Time since request: 18.962000 milliseconds]
    [Request URI: /]
    [Full request URI: http://www.people.com.cn/]
```

7 HTTP HTTP “ ->HTTP Stream”



8 ncat HTTP ncat HTTP

HTML

```
C:\Users\OMEN\Desktop\    \    \        \nmap-7.91\nmap-7.91>ncat  -C  www.xjtu.edu.cn  80
GET / HTTP/1.1
Host: www.xjtu.edu.cn
User-Agent: Zhang Zhaoyuan
Connection: close

HTTP/1.1 200 OK
Date: Wed, 18 Mar 2026 13:24:55 GMT
Server: **********
X-Frame-Options: SAMEORIGIN
Upgrade: h2c,h2
Connection: Upgrade, close
Last-Modified: Wed, 18 Mar 2026 12:28:05 GMT
Accept-Ranges: bytes
Content-Length: 57957
Cache-Control: max-age=600
Expires: Wed, 18 Mar 2026 13:34:55 GMT
Vary: Accept-Encoding
ETag: "e265-64d4b957a88ff-gzip"
Content-Type: text/html
Content-Language: zh-CN

 ?!DOCTYPE  HTML>
<HTML><HEAD><TITLE>        €     ?/TITLE><META  Name="keywords"  Content="        €     ?  />



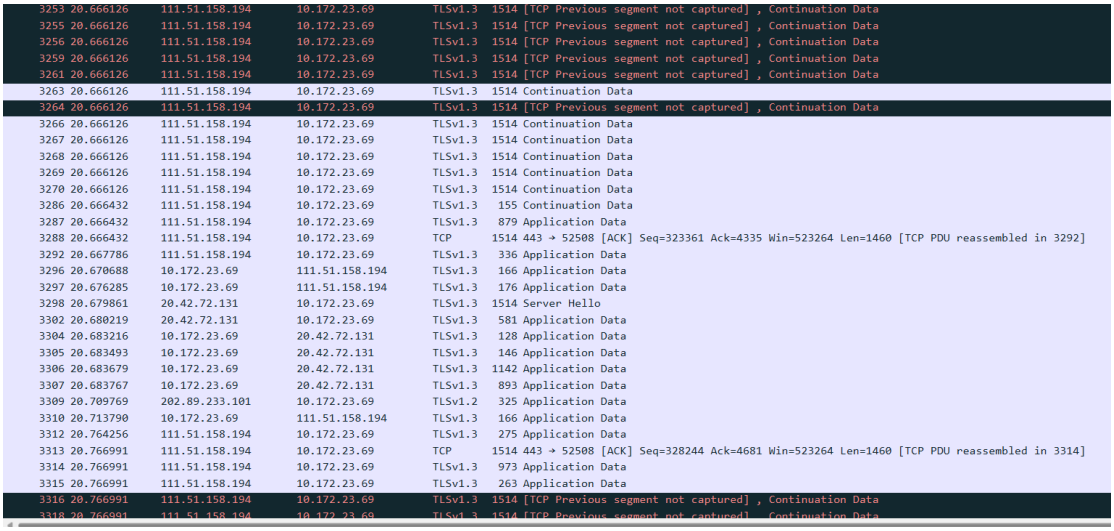

<META name="360-site-verification" content="845cb73defc117caad1186ca8fac8532"><script type="text/javascript">

if(/AppleWebKit.*Mobile/i.test(navigator.userAgent) ||(/MIDP|SymbianOS|NOKIA|SAMSUNG|LG|NEC|TCL|Alcatel|
BIRD|DBTEL|Dopod|PHILIPS|HAIER|LENOVO|MOT-|Nokia|SonyEricsson|SIE-|Amoi|
ZTE/.test(navigator.userAgent))){
      if(window.location.href.indexOf("?mobile")<0){
            try{
                  if(/Android|Windows Phone|webOS|iPhone|iPod|BlackBerry/i.test(navigator.userAgent)){
                        window.location.href="http://mob.xjtu.edu.cn/";
                  }
            }catch(e){}
      }
}

</script>
<META content="text/html; charset=UTF-8" http-equiv="Content-Type">
<META  content="IE=edge,chrome=1"  http-equiv="X-UA-Compatible"><LINK  rel="stylesheet"  type="text/css"
href="style/xjnew611.css"><script type="text/javascript" src="js/jquery.min.js"></script><script type="text/javascript"
src="js/jquery.SuperSlide.js"></script>
<script type="text/javascript" src="js/ddsmoothmenu.js"></script>
<META name="baidu-site-verification" content="r23jsHKdp1">
<STYLE>
.menulink ul li:hover ul{width: 230px !important;}
</STYLE>
```

9 HTTPS HTTPS

TLS TLS



2.3.3 HTTP

10 HTTP DNS TCP HTTP

10.1

```
import socket
import sys
import time

host = sys.argv[1]
port = 80

# DNS
dns_start = time.perf_counter()
addr_info = socket.getaddrinfo(host, port, socket.AF_INET, socket.SOCK_STREAM)
dns_end = time.perf_counter()

ip = addr_info[0][4][0]

#       TCP
sock = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

tcp_start = time.perf_counter()
sock.connect((ip, port))
tcp_end = time.perf_counter()

#
lines = []
while True:
    line = input()
    if line == "":
        break
    lines.append(line)

request = "\r\n".join(lines) + "\r\n\r\n"

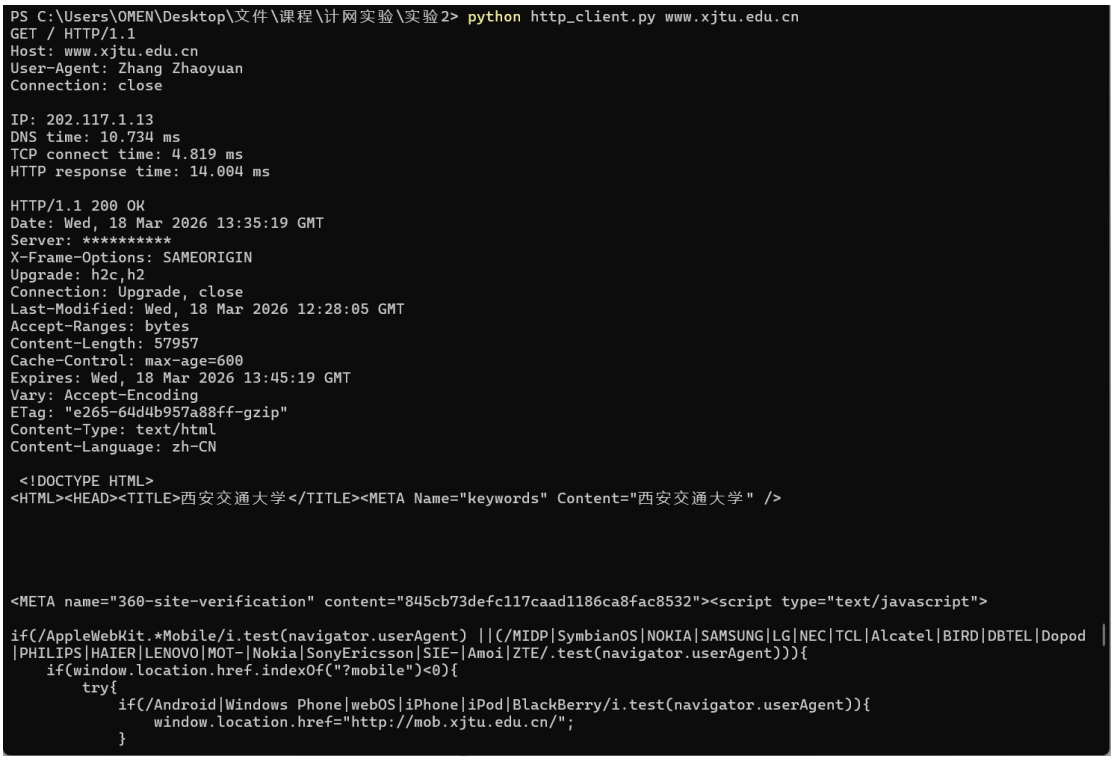
#
http_start = time.perf_counter()
sock.sendall(request.encode())

response = b""
while True:
    data = sock.recv(4096)
    if not data:
        break
    response += data
http_end = time.perf_counter()

sock.close()
```

```
print(f"IP: {ip}")
print(f"DNS time: {(dns_end - dns_start) * 1000:.3f} ms")
print(f"TCP connect time: {(tcp_end - tcp_start) * 1000:.3f} ms")
print(f"HTTP response time: {(http_end - http_start) * 1000:.3f} ms")
print()
print(response.decode(errors="ignore"))
```

10.2



10.3 DNS TCP HTTP ms

DNS	10.734 ms
TCP	4.819 ms
HTTP	14.004 ms

11 HTTP DNS TCP HTTP

11.1

```
import socket
```

```
import sys
import time
from html.parser import HTMLParser
from urllib.parse import urlparse, urljoin

USER_AGENT = "Zhang Zhaoyuan"

class EmbeddedObjectParser(HTMLParser):
    def __init__(self):
        super().__init__()
        self.urls = []

    def handle_starttag(self, tag, attrs):
        attrs = dict(attrs)

        if tag == "img" and "src" in attrs:
            self.urls.append(attrs["src"])

        elif tag == "script" and "src" in attrs:
            self.urls.append(attrs["src"])

        elif tag == "iframe" and "src" in attrs:
            self.urls.append(attrs["src"])

        elif tag == "source" and "src" in attrs:
            self.urls.append(attrs["src"])

        elif tag == "link":
            rel = attrs.get("rel", "").lower()
            href = attrs.get("href")
            if href and "stylesheet" in rel:
                self.urls.append(href)

def recv_all(sock):
    data = b""
    while True:
        part = sock.recv(4096)
        if not part:
            break
        data += part
    return data

def parse_headers(header_bytes):
    lines = header_bytes.decode("iso-8859-1").split("\r\n")
    status_line = lines[0]
    headers = {}

    for line in lines[1:]:
        if ": " in line:
            k, v = line.split(": ", 1)
            headers[k.lower()] = v

    parts = status_line.split(" ", 2)
    status_code = int(parts[1]) if len(parts) >= 2 else 0
    return status_code, headers

def decode_chunked(body):
    result = b""
    i = 0
    while True:
        j = body.find(b"\r\n", i)
        size_line = body[i:j]
        size = int(size_line.decode().split(";", 1)[0], 16)
        if size == 0:
            break
        i = j + 2
        result += body[i:i + size]
        i += size + 2
    return result

def fetch_once(url):
    parsed = urlparse(url)
    host = parsed.hostname
    port = parsed.port or 80
    path = parsed.path if parsed.path else "/"
```

```
    if parsed.query:
        path += "?" + parsed.query

    dns_start = time.perf_counter()
    addr_info = socket.getaddrinfo(host, port, socket.AF_INET, socket.SOCK_STREAM)
    dns_end = time.perf_counter()

    ip = addr_info[0][4][0]

    sock = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

    tcp_start = time.perf_counter()
    sock.connect((ip, port))
    tcp_end = time.perf_counter()

    host_header = host if port == 80 else f"{host}:{port}"
    request = (
        f"GET {path} HTTP/1.1\r\n"
        f"Host: {host_header}\r\n"
        f"User-Agent: {USER_AGENT}\r\n"
        f"Accept-Encoding: identity\r\n"
        f"Connection: close\r\n"
        f"\r\n"
    )

    http_start = time.perf_counter()
    sock.sendall(request.encode())
    raw_response = recv_all(sock)
    http_end = time.perf_counter()

    sock.close()

    header_bytes, body = raw_response.split(b"\r\n\r\n", 1)
    status_code, headers = parse_headers(header_bytes)

    if headers.get("transfer-encoding", "").lower() == "chunked":
        body = decode_chunked(body)

    return {
        "url": url,
        "host": host,
        "ip": ip,
        "status": status_code,
        "headers": headers,
        "body": body,
        "dns_ms": (dns_end - dns_start) * 1000,
        "tcp_ms": (tcp_end - tcp_start) * 1000,
        "http_ms": (http_end - http_start) * 1000,
    }

def fetch_with_redirect(url, max_redirect=3):
    records = []
    current_url = url

    for _ in range(max_redirect + 1):
        result = fetch_once(current_url)
        records.append(result)

        if result["status"] in (301, 302, 303, 307, 308):
            location = result["headers"].get("location")
            if not location:
                break
            next_url = urljoin(current_url, location)
            if not next_url.startswith("http://"):
                break
            current_url = next_url
        else:
            break

    return records

def unique_keep_order(urls):
    seen = set()
    result = []
    for u in urls:
```

```
            if u not in seen:
                seen.add(u)
                result.append(u)
    return result

def main():
    target = sys.argv[1]

    if not target.startswith("http://"):
        target = "http://" + target

    all_records = []

    #
    main_records = fetch_with_redirect(target)
    all_records.extend(main_records)
    final_main = main_records[-1]

    html = final_main["body"].decode(errors="ignore")

    #
    parser = EmbeddedObjectParser()
    parser.feed(html)

    object_urls = []
    for u in parser.urls:
        abs_url = urljoin(final_main["url"], u)
        if abs_url.startswith("http://"):
            object_urls.append(abs_url)

    object_urls = unique_keep_order(object_urls)

    #
    for obj_url in object_urls:
        obj_records = fetch_with_redirect(obj_url)
        all_records.extend(obj_records)

    #
    print("===== Download Result =====")
    print(f"Main Page: {final_main['url']}")
    print(f"Embedded Object Count: {len(object_urls)}")
    print()

    total_dns = 0.0
    total_tcp = 0.0
    total_http = 0.0

    for i, r in enumerate(all_records, 1):
        total_dns += r["dns_ms"]
        total_tcp += r["tcp_ms"]
        total_http += r["http_ms"]

        print(f"[{i}] {r['url']}")
        print(f"     IP: {r['ip']}")
        print(f"     Status: {r['status']}")
        print(f"     DNS time: {r['dns_ms']:.3f} ms")
        print(f"     TCP connect time: {r['tcp_ms']:.3f} ms")
        print(f"     Full HTTP response time: {r['http_ms']:.3f} ms")
        print()

    print("===== Total =====")
    print(f"Total DNS time: {total_dns:.3f} ms")
    print(f"Total TCP connect time: {total_tcp:.3f} ms")
    print(f"Total Full HTTP response time: {total_http:.3f} ms")

if __name__ == "__main__":
    main()
```

11.2

```
PS C:\Users\OMEN\Desktop\文件\课程\计网实验\实验2> python http_client.py www.baidu.com
===== Download Result =====
Main Page: http://www.baidu.com
Embedded Object Count: 1

[1] http://www.baidu.com
    IP: 36.152.44.93
    Status: 200
    DNS time: 18.238 ms
    TCP connect time: 26.522 ms
    Full HTTP response time: 29.981 ms

[2] http://www.baidu.com/img/flexible/logo/pc/index.png
    IP: 36.152.44.93
    Status: 200
    DNS time: 0.367 ms
    TCP connect time: 26.942 ms
    Full HTTP response time: 31.587 ms

===== Total =====
Total DNS time: 18.605 ms
Total TCP connect time: 53.464 ms
Total Full HTTP response time: 61.568 ms
```

11.3 DNS TCP HTTP ms

DNS	18.605
TCP	2
TCP	53.464
HTTP	61.568